DAY — 10 SEAT NUMBER

2011	III	11	1100	J - 585	(E)

CHEMISTRY PAPER - I (55)

(PHYSICAL AND INORGANIC CHEMISTRY)

Time : 2 Hrs. (4 Pages) Max. Marks : 40

Note : *(i) All questions carry equal marks.*

(ii) Give balanced equations and draw neat diagrams wherever necessary.

(iii) Use of logarithmic table is allowed.

(iv) Figures to the right hand side indicate full marks.

(v) Answer to every question must be written on a new page.

Q. 1. Select and write the most appropriate answer from the given alternatives for each sub-question : [8]

(i) The heats of reaction at constant pressure and constant volume are related by (1)

(a) $\Delta H - \Delta E = \Delta nRT$ (b) $\Delta H + \Delta E = \Delta nRT$

(c) $\Delta H = \Delta E$ (d) $\Delta H = \frac{\Delta E}{\Delta nRT}$

(ii) Which of the following can act as both, conjugate acid and conjugate base? (1)

(a) CO_3^{--} (b) H_2CO_3

(c) HCO_3^- (d) H_3O^+

(iii) Zinc is used to protect iron from corrosion, because (1)

(a) Zinc is easily available.

(b) E°_{Red} of Zn is less than E°_{Red} of Fe.

(c) E°_{Red} of Zn is greater than E°_{Red} of Fe.

(d) Zinc is an inert metal.

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(iv) The particles given out during artificial transmutation are called (1)

(a) Target (b) Recoiled nucleus

(c) Projectiles (d) Emission

(v) Unit of rate constant of a first order reaction is (1)

(a) $mol\ dm^{-3}\ S^{+1}$ (b) S^{-1}

(c) $mol\ dm^{+3}\ S^{-1}$ (d) S^{+1}

(vi) The colour of transition metal ions is due to (1)

(a) small size of metal cations (b) completely filled ns- orbital

(c) d – d transition of electrons (d) absorption of light in invisible region

(vii) The molarity and normality of a solution containing $4{\cdot}9 \times 10^{-3}$ kg of H_2SO_4 per dm^3 of the solution respectively are (1)

(a) 0·05 M and 0·1 N (b) 0·1 M and 0·05 N

(c) 0·5 M and 1 N (d) 1·0 M and 0·5 N

(viii) An ideal gas expands by a volume of 2 dm^3 at 300 K against a constant pressure of $1{\cdot}013 \times 10^5$ N/m^2. The work obtained is (1)

(a) – 202·6 kJ (b) – 202·6 J

(c) +202·6 J (d) +202·6 kJ

Q. 2. (A) Attempt any ONE : [8]

(i) Distinguish between order of Reversible process and Irreversible process. (2)

(ii) Derive van't Hoff equation for solution. (2)

(B) Attempt any ONE :

(i) Define : (a) Hydrolysis of salt

(b) Faraday (2)

(ii) Distinguish between order of molecularity and order of reaction. (2)

(C) Answer the following :

(i) Transition elements show variable oxidation states. Explain. (2)

(ii) Define a Solution. Explain the terms Solute and Solvent. (2)

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Q. 3. (A) Attempt any ONE : [8]

(i) Describe the determination of molecular weight from lowering of vapour pressure by Ostwald-Walker method. (3)

(ii) Define Radioactivity. Write two applications of radioisotopes in (a) Medicine, (b) Agriculture. (3)

(B) Attempt any ONE :

(i) Derive an integrated rate equation for first order reaction. (3)

(ii) What are Transition Elements? Explain their position in Modern Periodic Table and write the electronic configuration of Co. (At. No. 27) (3)

(C) Answer the following :

What is Corrosion? Explain the need for prevention of corrosion. (2)

Q. 4. (A) Answer the following : [8]

Explain the concept of Maximum Work. Derive an expression for maximum work done in isothermal reversible expansion of one mole of an ideal gas. (4)

(B) Answer any ONE :

(i) State and explain common ion effect. Explain its 'two' applications. (4)

(ii) Explain the construction and working of lead accumulator. (4)

Q. 5. (A) Attempt any ONE : [8]

(i) Calculate heat of formation of anhydrous $Al_2Cl_{6(s)}$ from the following data –

(1) $2Al_{(s)} + 6HCl_{(aq.)} \rightarrow Al_2Cl_{6(aq.)} + 3H_{2(g)}$; $\Delta H_1 = -1003$ kJ

(2) $Al_2Cl_{6(s)} + aq. \rightarrow Al_2Cl_{6(aq.)}$; $\Delta H_2 = -643$ kJ

(3) $H_{2(g)} + Cl_{2(g)} \rightarrow 2HCl_{(g)}$; $\Delta H_3 = -184$ kJ

(4) $HCl_{(g)} + aq. \rightarrow HCl_{(aq.)}$; $\Delta H_4 = -73$ kJ (4)

(ii) The degree of dissociation of a monoacidic base is 2% in its centimolar solution. Calculate its degree of dissociation, percent degree of dissociation and pOH of the solution in its milimolar solution. (4)

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(B) Attempt any TWO :

(i) The half life period of radioactive carbon ${}^{14}_{6}C$ isotope is 5770 years. Calculate its decay constant and time required for 70% disintegration of it. (2)

(ii) Write the Net Cell Reaction and calculate e. m. f . of the following cell at 298 K.

$$Al_{(s)} \left| Al^{3+}_{(aq.)} \atop 1M \right| \left| \left| Cu^{2+}_{(aq.)} \atop 1M \right| Cu_{(s)}$$

(Given : $E^{\circ}_{Al_{(Red)}} = -1{\cdot}66V$, $E^{\circ}_{Cu_{(Red)}} = +0{\cdot}34V$) (2)

(iii) Calculate the osmotic pressure of a solution of urea containing 5 gram of it in 100 ml of solution at 273 K.

(Given : Molecular weight of urea = 60 and

$R = 0{\cdot}082\ dm^3 - atm\ mol^{-1}\ K^{-1}$) (2)

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